Algebra Exponents & Exponential Functions Cheat Sheet

Unit Focus: *exponent rules and exponential growth/decay*

1 KEY VOCABULARY

base	The number being multiplied repeatedly.	In 3^4, the base is 3.
exponent	Shows how many times the base is used as a factor.	In 3^4, the exponent is 4.
power	A number written using a base and an exponent.	3^4 is a power.
expanded form	A product that shows the base used as a factor.	$3^4 = 3 \cdot 3 \cdot 3 \cdot 3$
exponential form	A compact way to write a repeated product.	$3 \cdot 3 \cdot 3 \cdot 3 = 3^4$
exponential function	A function where the variable is in the exponent.	$f(x) = 2^x$
growth factor	The constant multiplier in exponential growth (where the factor is greater than 1).	In $f(x) = 3 \cdot 1.5^x$, the growth factor is 1.5.
decay factor	The constant multiplier in exponential decay (where the factor is between 0 and 1).	In $f(x) = 4 \cdot 0.7^x$, the decay factor is 0.7.

2 EVALUATING POWERS

Find the value of each power. Remember to use parentheses when the base is negative.

Example 1: Evaluate 2^4.

Step 1: Write as a product: $2^4 = 2 \cdot 2 \cdot 2 \cdot 2$

Step 2: Multiply: $2^4 = 16$

Answer: 16

Example 2: Evaluate $(-3)^2$ and -3^2.

With parentheses (negative number squared):

$(-3)^2 = (-3)(-3)$

$= 9$

Answer: 9

Without parentheses (exponent only on 3):

$-3^2 = -(3^2)$

$= -9$

Answer: −9

3 PRODUCT OF POWERS

When multiplying powers with the same base, add the exponents.

Rule: $a^m \cdot a^n = a^{m+n}$

Example: Simplify $x^3 \cdot x^5$.

Step 1: Add the exponents: $3 + 5 = 8$

Step 2: Write the result: $x^3 \cdot x^5 = x^8$

Answer: x^8

4 QUOTIENT OF POWERS

When dividing powers with the same base, subtract the exponents ($a \neq 0$).

Rule: $\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0)$

Example: Simplify $\frac{x^7}{x^2}$.

Step 1: Subtract the exponents: $7 - 2 = 5$

Step 2: Write the result: $\frac{x^7}{x^2} = x^5$

Answer: x^5

5 POWER OF A POWER

When a power is raised to another power, multiply the exponents.

Rule: $(a^m)^n = a^{mn}$

Example: Simplify $(x^3)^4$.

Step 1: Multiply the exponents: $3 \cdot 4 = 12$

Step 2: Write the result: $(x^3)^4 = x^{12}$

Answer: x^{12}

6 POWER OF A PRODUCT

When a product is raised to a power, raise each factor to that power.

Rule: $(ab)^n = a^n b^n$

Example: Simplify $(2x)^3$.

Step 1: Raise each factor to the power: $(2x)^3 = 2^3 x^3$

Step 2: Simplify: $(2x)^3 = 8x^3$

Answer: $8x^3$

7 ZERO AND NEGATIVE EXPONENTS

Use these rules for zero and negative exponents ($a \neq 0$).

Zero Exponent Rule:

$a^0 = 1 \quad (a \neq 0)$

Example: Evaluate 5^0.

$5^0 = 1$

Answer: 1

Negative Exponent Rule:

$a^{-n} = \frac{1}{a^n} \quad (a \neq 0)$

Example: Evaluate 2^{-3}.

$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$

Answer: $\frac{1}{8}$

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Add exponents only when multiplying like bases.
- Subtract exponents only when dividing like bases.
- A negative exponent means take the reciprocal, not a negative value.
- Parentheses matter, especially with negative bases.

COMMON MISTAKES:

- Adding or subtracting exponents when the bases are different.
- Thinking a^{-n} means $-a^n$.
- Forgetting parentheses with negative bases (e.g., $-3^2 \neq (-3)^2$).
- Not applying the power to every factor in a product (e.g., $(2x)^3 \neq 2x^3$).
- Forgetting that $a^0 = 1$ only works for $a \neq 0$.

Front

Algebra Exponents & Exponential Functions Cheat Sheet

More Practice: *scientific notation and exponential functions*

1 SCIENTIFIC NOTATION

Scientific notation writes a very large or very small number as a **product:** $a \times 10^n$

- $1 \le a < 10$ (the coefficient a has one nonzero digit to the left of the decimal).
- n is an integer (can be positive or negative).

Examples:

a) Write 4,500,000 in scientific notation.

Step 1: Move the decimal 6 places to the left.

Step 2: Write using a power of 10.

$4{,}500{,}000 = 4.5 \times 10^6$

Answer: 4.5×10^6

b) Write 0.00032 in scientific notation.

Step 1: Move the decimal 4 places to the right.

Step 2: Write using a power of 10.

$0.00032 = 3.2 \times 10^{-4}$

Answer: 3.2×10^{-4}

2 OPERATIONS WITH SCIENTIFIC NOTATION

To multiply or divide numbers in scientific notation, multiply or divide the coefficients and add or subtract the exponents of 10.

Multiplication Example:

Multiply $(3.2 \times 10^4)(5.0 \times 10^3)$.

Step 1: Multiply coefficients: $3.2 \times 5.0 = 16$.

Step 2: Add exponents: $10^4 \times 10^3 = 10^{4+3} = 10^7$.

Step 3: Write in scientific notation ($1 \le a < 10$): $16 \times 10^7 = 1.6 \times 10^8$.

Answer: 1.6×10^8

Division Example:

Divide $(6.0 \times 10^7) \div (2.4 \times 10^3)$.

Step 1: Divide coefficients: $6.0 \div 2.4 = 2.5$.

Step 2: Subtract exponents: $10^7 \div 10^3 = 10^{7-3} = 10^4$.

Step 3: Write in scientific notation: 2.5×10^4.

Answer: 2.5×10^4

3 EXPONENTIAL FUNCTIONS

An **exponential function** has the form $y = a(b^x)$.

a	The **initial value** (the value of y when $x = 0$).
b	The **growth or decay factor** (how much y is multiplied by each time x increases by 1).
x	The **independent variable** (often represents time).
y	The **dependent variable** (the output value).

4 GROWTH VS. DECAY

The value of the base b tells us whether the function models growth or **decay**.

Exponential Growth

If $b > 1$, the function models **exponential growth.** The value increases by a constant factor each time x increases by 1.



Example: $y = 3(1.5^x)$

Exponential Decay

If $0 < b < 1$, the function models **exponential decay.** The value decreases by a constant factor each time x increases by 1.



Example: $y = 8(0.6^x)$

5 WRITING AN EXPONENTIAL FUNCTION FROM A TABLE

Find an exponential function that fits the table.

x	0	1	2	3
y	5	10	20	40

Step 1: Identify the initial value. When $x = 0$, $y = 5$, so $a = 5$.

Step 2: Find the common ratio. $10 \div 5 = 2$, $20 \div 10 = 2$, $40 \div 20 = 2$. The outputs are multiplied by 2 each time.

Step 3: Write the function. $a = 5$ and $b = 2$, so $y = 5(2^x)$

6 EVALUATING AN EXPONENTIAL FUNCTION

Find the value of the function when x is given.

Example: For $y = 3(2^x)$, find y when $x = 4$.

Step 1: Substitute $x = 4$. $y = 3(2^4)$

Step 2: Evaluate the exponent. $y = 3(16)$

Step 3: Multiply. $y = 48$

Answer: $y = 48$

7 REAL-WORLD EXAMPLE

A population of 200 people grows by 10% each year. Write an exponential function and find the population after 3 years.

Step 1: Write the function.

Initial value: 200

Growth factor: $1 + 0.10 = 1.10$

$y = 200(1.10^x)$

Step 2: Find y when $x = 3$.

$y = 200(1.10^3)$

$= 200(1.331)$

$= 266.2$

Answer: 266.2 people

8 QUICK REMINDERS & STRATEGY TIPS

- **Initial value:** The coefficient a is the value when $x = 0$.
- **Growth or decay factor:** The base b tells how much the output is multiplied by each step.
- **Repeated multiplication:** In an exponential function, each increase of 1 in x multiplies the previous value by b.
- **Check for a constant ratio:** When given a table, divide consecutive y-values. If the ratio is constant, it's exponential.
- **Use scientific notation:** Helpful for very large or very small numbers to make calculations easier.